



Introduction

- Radar Cross Section (RCS) is one of the most important parameters determining the detectability of an aircraft by radar systems. While conventional aircraft are primarily designed for aerodynamic efficiency, modern stealth aircraft incorporate geometric shaping and electromagnetic design principles to reduce radar visibility.
- This project presents a computational framework developed entirely in Python for predicting the RCS of aircraft using high-frequency electromagnetic techniques. The work investigates how aircraft geometry influences scattering behavior and compares conventional and stealth configurations over multiple viewing angles and frequencies.

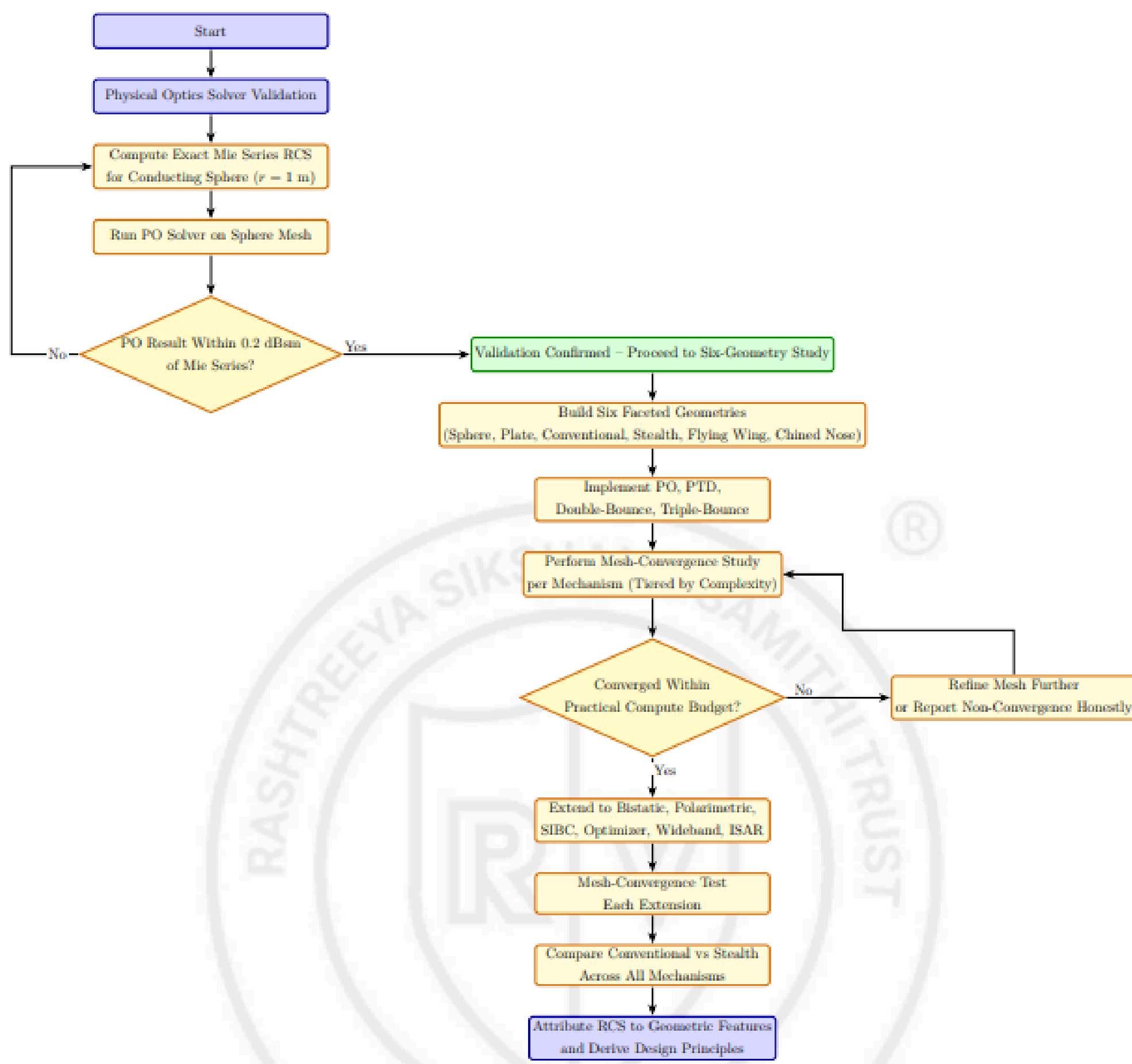
Problem Definition

Accurate prediction of Radar Cross Section generally requires computationally expensive electromagnetic solvers. Commercial software is often inaccessible for academic use and provides limited insight into the underlying scattering mechanisms. This project develops a first-principles numerical framework capable of predicting aircraft RCS while identifying the physical contributions of specular reflection, edge diffraction and multiple reflections. The developed tool also enables comparative analysis of conventional and stealth aircraft configurations.

Objectives

- Develop a first-principles Python framework for RCS prediction.
- Implement Physical Optics (PO) and Physical Theory of Diffraction (PTD).
- Model double-bounce and triple bounce scattering mechanisms.
- Validate the solver using canonical electromagnetic targets.
- Compare radar signatures of conventional and stealth aircraft.
- Investigate the influence of geometry on radar visibility with geometry optimization

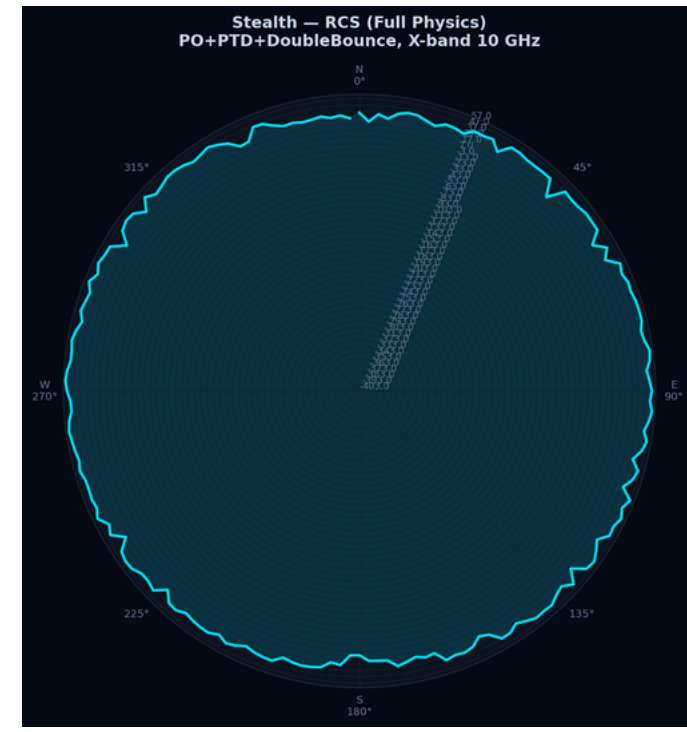
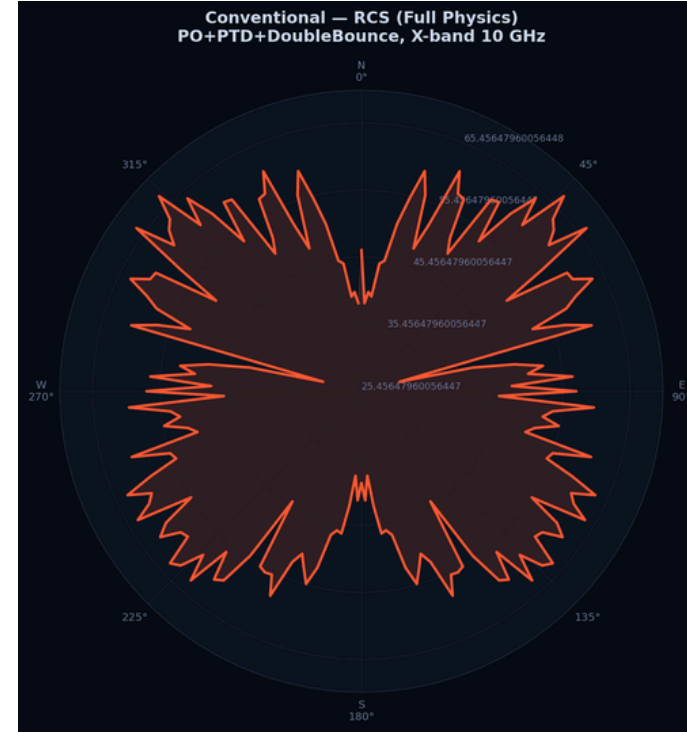
Methodology



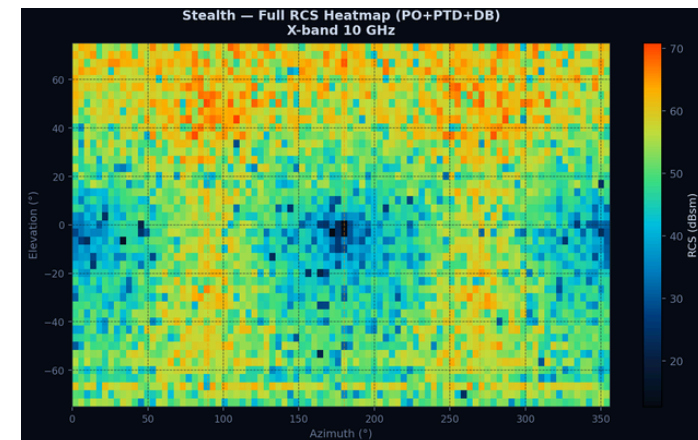
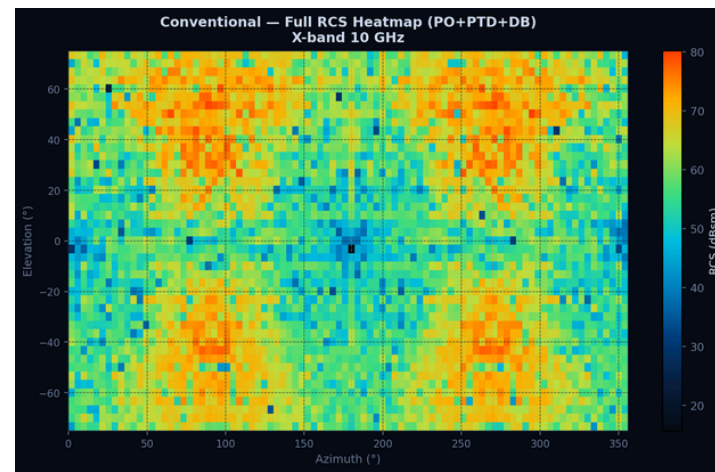
Tools used

- Python - used to model the entire project with first principles
- NumPy - library in python used for array creation
- SciPy - used for all mathematical calculations
- Matplotlib - used for plotting

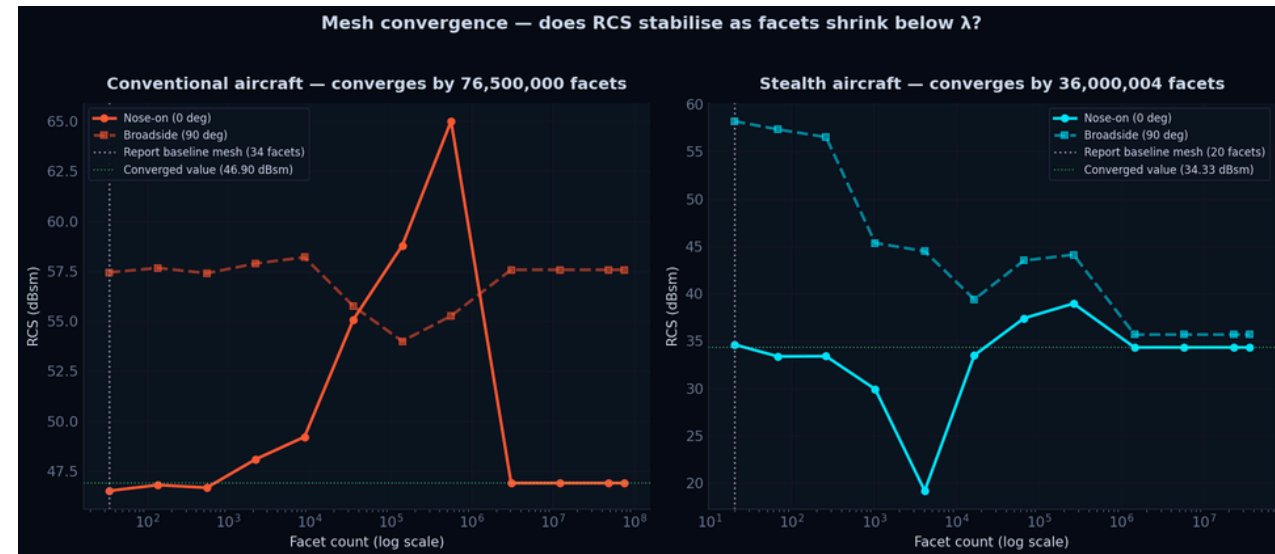
Results and Discussions



The stealth configuration exhibits significantly lower radar returns across most viewing angles due to optimized geometric shaping.



Full azimuth-elevation heatmaps reveal strong reflections from conventional aircraft while the stealth geometry suppresses high-intensity scattering regions.



Mesh convergence study

Conclusions

- Successfully developed a complete first-principles RCS prediction framework in Python.
- Numerical predictions were validated using canonical electromagnetic solutions.
- Stealth aircraft exhibited substantially reduced radar signatures compared with conventional configurations.
- Geometric shaping proved to be the primary factor governing RCS reduction.
- The developed framework can serve as an educational and research platform for computational electromagnetics.

Outcome of the work

- Developed an interactive computational RCS analysis platform.
- Implemented multiple high-frequency scattering models from first principles.
- Generated comparative RCS databases for conventional and stealth aircraft.
- Established a reusable workflow for future stealth design investigations.
- Improved understanding of electromagnetic scattering in aerospace applications.

References

- Knott, Radar Cross Section
- Balanis, Advanced Engineering Electromagnetics
- Skolnik, Introduction to Radar Systems
- Ufimtsev, Physical Theory of Diffraction
- Harrington, Field Computation by Moment Methods